BLOCKCHAIN-BASED INTERNSHIP VERIFICATION

ProofPass

Issue once. Verify anywhere. Trust the record.



Internship proof has a trust gap.

Claims move quickly. Verification still depends on slow, fragmented evidence.

01

A claim is easy to make

Profiles, résumés and screenshots can be edited, duplicated or stripped of context.

TRUST GAP



02

Proof is hard to check

Recruiters must contact issuers, inspect documents and trust disconnected databases.

One record. Three trusted roles.

ProofPass turns an internship credential into a controlled, tamper-evident public record.



Approved organizations create records.

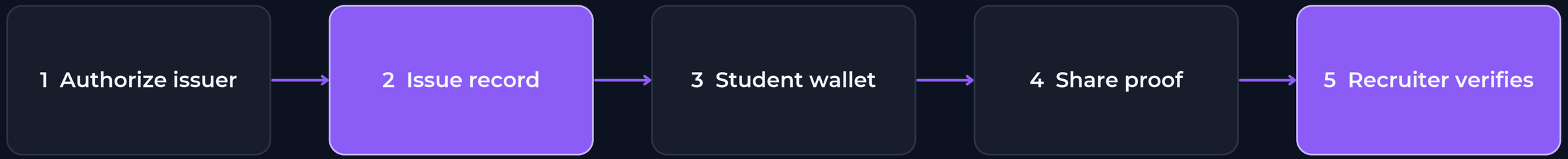
Students receive a durable credential reference.

Anyone can validate status and issuer on-chain.

The blockchain anchors provenance — the interface makes it usable.

How ProofPass works

A complete credential path with no private verification portal required.



ProofPass.sol stores issuer authority, credential identity and validity state. The wallet carries the reference; the chain remains the source of truth.

Core features

A focused DApp for issuing, carrying and checking internship credentials.

01

Issuer authorization

Owner-controlled issuer registry.

02

Credential issuance

Structured, unique on-chain records.

03

Wallet connection

Role-aware actions via browser wallet.

04

Public verification

Lookup by credential identifier.

05

Revocation status

Current validity is explicit and auditable.

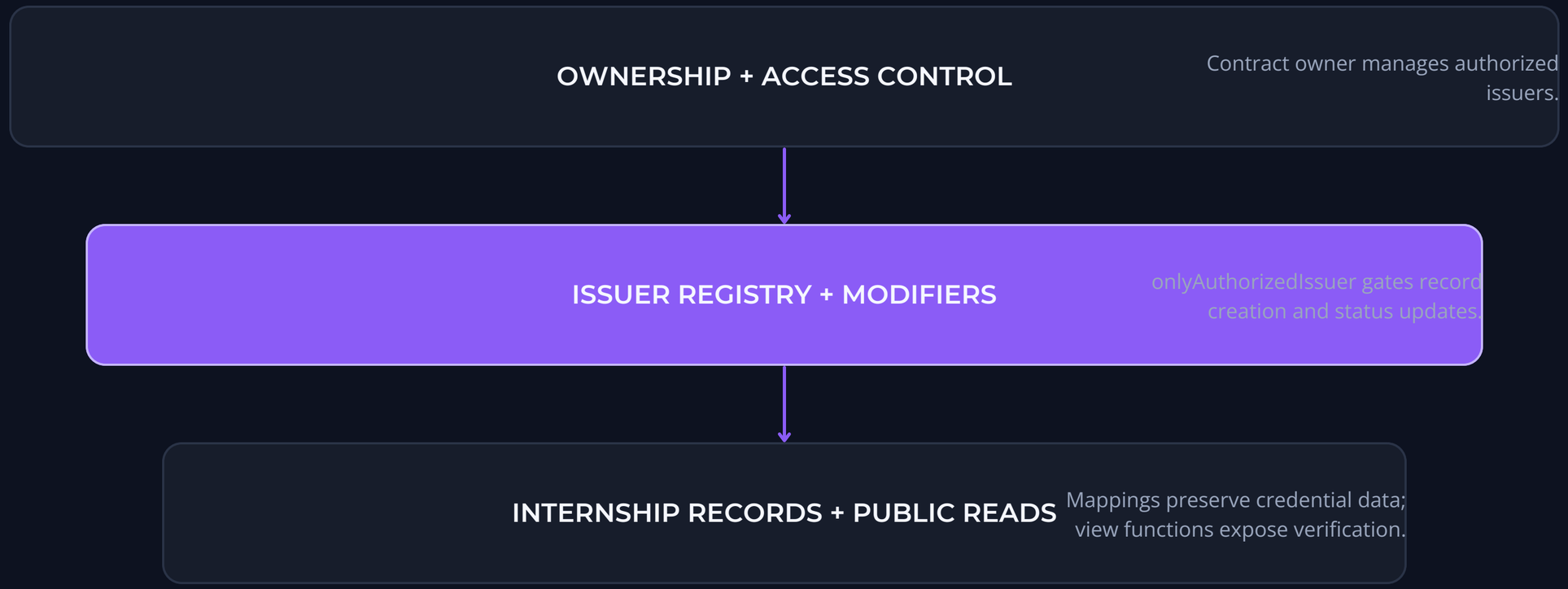
06

Shareable proof

Copyable link and QR-ready flow.

Smart contract architecture

ProofPass.sol separates authority, data and read access.



Technology stack

Four layers, each chosen for a clear responsibility.

Frontend	React • Vite • ethers.js
Wallet	Browser wallet • account signing
Blockchain	EVM test network • RPC provider
Contract	Solidity • events • mappings

DApp walkthrough

The frontend reveals the right action for the connected role.

01 Connect wallet

02 Select role-aware action

03 Enter credential details

04 Confirm transaction



Verification in seconds

A recruiter needs only a shared link, QR code or credential ID.

01

Open

Scan QR or follow the shareable link.

02

Resolve

Read the credential directly from the contract.

03

Confirm

See issuer, student, dates and active status.



Controlled records. Meaningful verification.

Validation happens before state changes — and remains visible afterward.

✓ Only owner authorizes issuers

✓ Only issuers create records

✓ Credential IDs stay unique

✓ Required fields cannot be empty

✓ Dates follow valid ordering

✓ Revocation preserves history

Testing & results

Functional tests were run after deployment against expected roles and failure states.

Owner authorizes issuer	PASS
Issuer creates credential	PASS
Public lookup returns record	PASS
Duplicate ID is rejected	PASS
Unauthorized issue is rejected	PASS
Revoked record shows inactive	PASS

6 / 6 functional paths passed

Proof that travels with the student.

ProofPass replaces unverifiable claims with transparent, issuer-backed records.

NOW

Value delivered

A credible end-to-end DApp: controlled issuance, student-friendly sharing and instant public verification.

NEXT

Future scope

Decentralized identity, privacy-preserving proofs, multi-chain deployment, richer issuer governance and institutional integrations.

LIVE DEMO

See ProofPass in action.

Connect → issue → share → verify

OPEN PROJECT DEMO

Repository • Contract • Frontend

